

MODULE 3

LEARNING PRESENTATION: MODELING MULTI-TABLE DATA RELATIONSHIPS

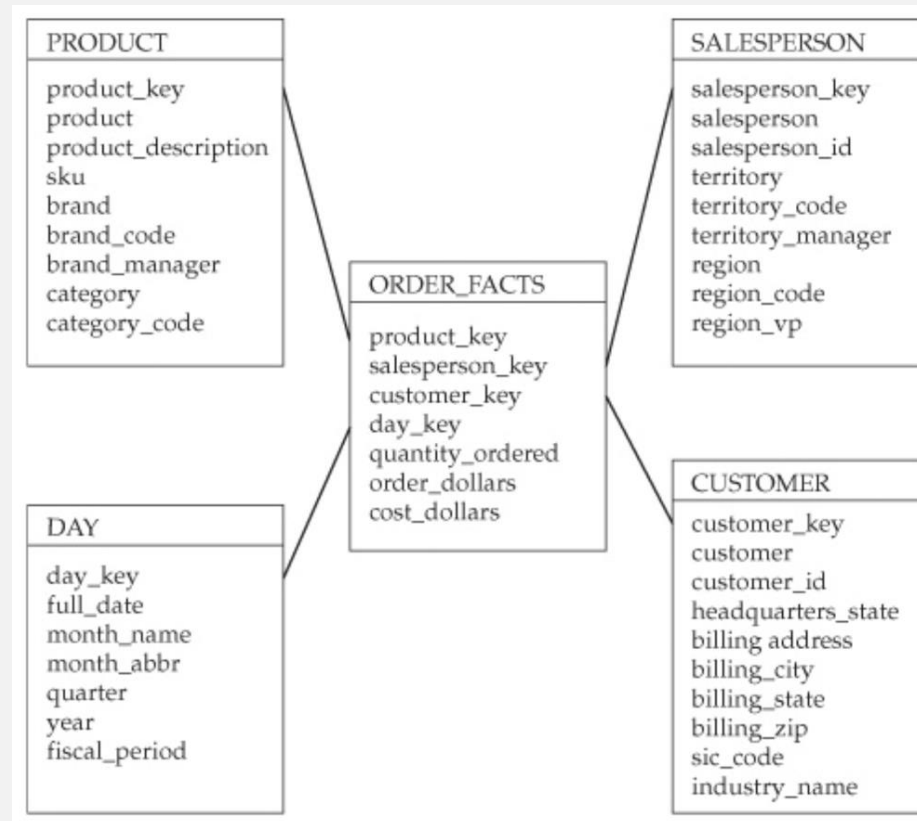
Scott Person

CONCEPTS

- tables
- table relationships
- joins
- primary and foreign keys
- star schema versus snowflake schema
- the meaning of "fact table" and "dimension table" in a star schema

STAR SCHEMA

- Starting with the end
- Simple, elegant, efficient
- Fact table one join away from each dimension table
- Primary advantage: clear query path for analytical tools



TABLES

- Organization of data
- Columns and Rows
- Primary Key

customer_id: 9900011
cust_name: Johnson, Sue
cust_birth_date: 3/2/1961
address_state: AZ
(more attributes) ...

Customer dimension table

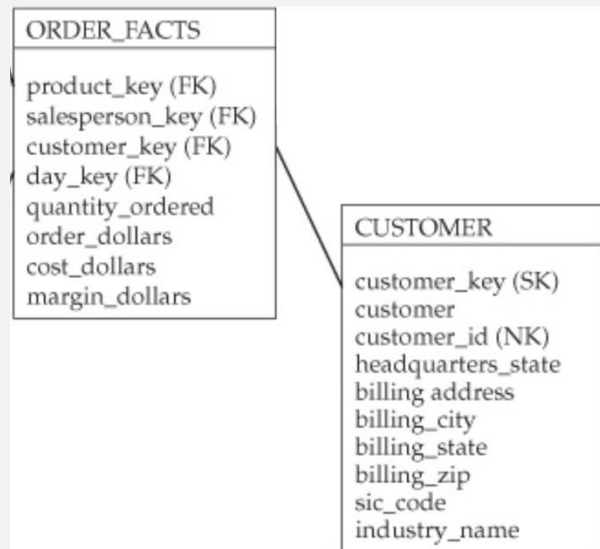
customer_ key	customer_ id	customer_ name	date_of_ birth	state	
1011	1140400	Davis, Robert	7/8/1932	NY	
1022	3305300	Nguyen, Tamara	1/2/1965	DC	
1302	7733300	Rodriguez, Jason	4/5/1972	MA	
1499	9900011	Johnson, Sue	3/2/1961	AZ	

More
attributes...

Record for Sue in
Dimension table

TABLE RELATIONSHIPS, JOINS, KEYS

- One to many cardinality
- Joins made on the keys
- Primary/Foreign key



Customer dimension table

customer_key	customer_id	customer_name	date_of_birth	state	
1011	1140400	Davis, Robert	7/8/1932	NY	
1022	3305300	Nguyen, Tamara	1/2/1965	DC	
1302	7733300	Rodriguez, Jason	4/5/1972	MA	
1499	9900011	Johnson, Sue	3/2/1961	AZ	

More attributes...

Record for Sue in Dimension table

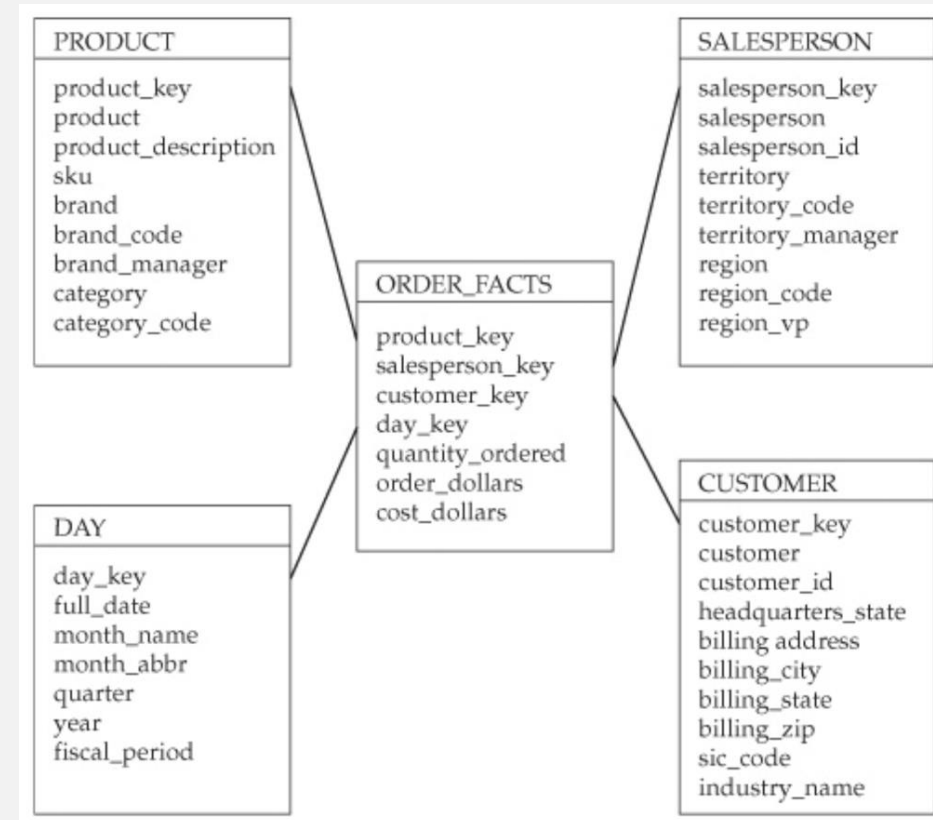
Orders fact table

customer_key	day_key	product_key	quantity_ordered	
1499	2322	10119	5	

The diagram shows a one-to-many relationship between the **Customer dimension table** and the **Orders fact table**. The **customer_key** (1499) in the **Orders fact table** is linked to the **customer_key** (1499) in the **Customer dimension table**, which corresponds to the record for Sue Johnson.

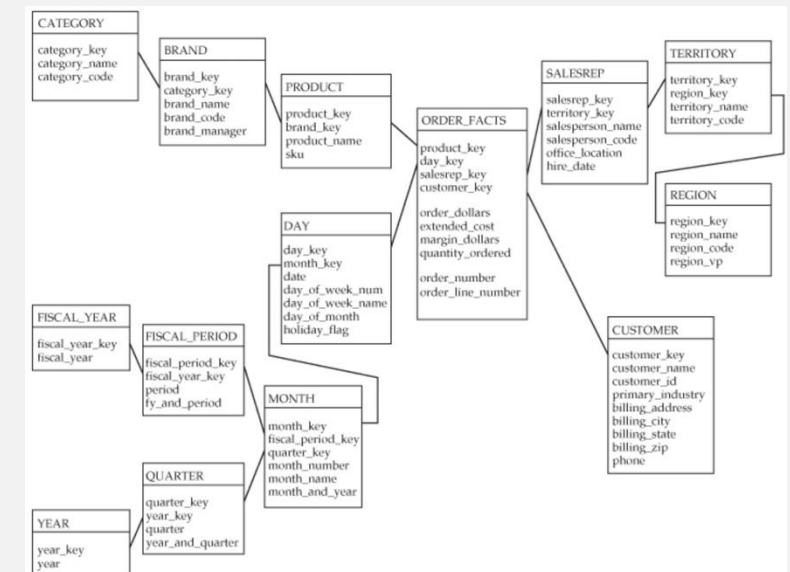
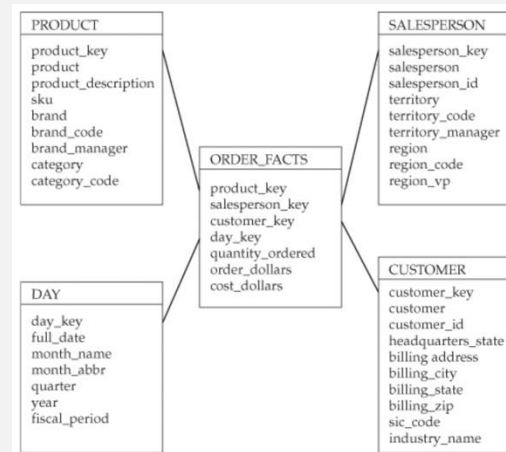
FACT AND DIMENSION TABLES

- Facts are events or Snapshots in time
- Dimensions provide context to the facts
- “By Rule” – Show me facts BY dimensions



STAR VS SNOWFLAKE SCHEMAS

- Snowflake schemas are seductive – nicely organized less data duplication
- In practice they have few benefits
- Increase data maintenance complexity
- Slower performance



CITATION

Adamson, C. (2010). *Star schema: The complete reference*. McGraw-Hill.

